

Week 3 - Part 3 - Osprey1st College Algebra - Summer 2024

Section covered: 3.3

Because the square of a real number is never negative, there is no real number x such that $x^2 = -1$. To provide a setting in which such equations have solutions, mathematicians invented an expanded system of numbers, the complex numbers. The **imaginary number i** , defined to be a solution of the equation $x^2 = -1$, is the basis of this new set.

The Imaginary Unit i

The **imaginary unit i** is defined as

$$i = \sqrt{-1}, \text{ where } i^2 = -1.$$

$$i^2 = -1$$

$$\sqrt{-25} = \sqrt{25} \sqrt{-1} = 5i$$

Complex Numbers and Imaginary Numbers

The set of all numbers in the form

$$a + bi$$

with real numbers a and b , and i , the imaginary unit, is called the set of **complex numbers**. The real number a is called the **real part** and the real number b is called the **imaginary part** of the complex number $a + bi$.

Examples of complex numbers:

$5 + 2i$ - real part 5 , imaginary part 2 ; nonreal complex number

$1 - 3i$ - real part 1 , imaginary part -3 ; nonreal complex number

$4i = 0 + 4i$ - real part 0 ; imaginary part 4 ; nonreal complex number called also pure imaginary number

$3 = 3 + 0i$ - real part 3 , imaginary part 0 ; real number.

Note: Real numbers are special cases of complex numbers, where imaginary part is equal to zero!

Square Root of a Negative Number

Consider the following roots:

$$\sqrt{-16} = \sqrt{(-1)16} = \sqrt{-1}\sqrt{16} = i \cdot 4 = 4i;$$

$$\sqrt{-25} = \sqrt{(-1)25} = \sqrt{-1}\sqrt{25} = i \cdot 5 = 5i$$

The Expression $\sqrt{-a}$

If a is a positive real number then

$$\sqrt{-a} = i\sqrt{a}$$

Problem 8. Write as the product of a real number and i , using the definition of $\sqrt{-a}$.

(a) $\sqrt{-81}$

$$9i$$

(b) $\sqrt{-98}$

$$7i\sqrt{2}$$

Adding and Subtracting Complex Numbers

Adding and Subtracting Complex Numbers

$$(a + bi) + (c + di) = (a + c) + (b + d)i \quad \text{and} \quad (a + bi) - (c + di) = (a - c) + (b - d)i$$

Note: Treat real parts as like terms and imaginary parts as like-terms.

Problem 9. Add or subtract two complex numbers.

(a) $*(2 + 5i) + (3 - 2i)$

$$5 - 3i$$

$a + bi$

(b) $*(-10 + 9i) - (-13 + 10i)$

$$+13 - 10i$$

$$3 - i$$

Multiplying Complex numbers

To multiply two complex numbers of the form $a + bi$, we multiply as though they are binomials. Then we use the relationship $i^2 = -1$ to simplify.

Problem 10. * Multiply the complex numbers. Write the product in the form $a + bi$.

(a) $-8i(2i - 7) = -16i^2 + 56i$
 $-16(-1)$
 $16 + 56i$

(b) $(-1 + 4i)(3 + i)$
 $-3 - i + 12i + 4i^2$
 $+4(-1)$
 $-3 + 11i - 4$
 $-7 + 11i$

(c) $(2 - 3i)^2$
 $4 - 6i - 6i + 9i^2$
 $+9(-1)$
 $4 - 12i - 9$
 $-5 - 12i$

(d) $(5 + 2i)(5 - 2i)$
 $25 - 10i + 10i - 4i^2$
 $-4(-1)$
 $25 + 4 = 29$

Dividing Complex Numbers

To divide complex numbers we need to multiply both the numerator and the denominator by the conjugate of the complex number in the denominator.

Conjugate of a Complex Number

The complex numbers $a + bi$ and $a - bi$ are called **complex conjugates** of each other. The multiplication of complex conjugates gives a real number.

$$(a + bi)(a - bi) = a^2 + b^2$$

Example:

The complex conjugate of $4 - 5i$ is $4 + 5i$. Let's multiply them.

$$(4 - 5i)(4 + 5i) = 16 + 20i - 20i - 25i^2 = 16 - 25i^2 = 16 - 25(-1) = 16 + 25 = 41$$

To divide complex numbers we **multiply both the numerator and the denominator by the complex conjugate of the denominator**. For example:

$$\frac{5i}{2 - i} = \frac{(5i)}{(2 - i)} \cdot \frac{(2 + i)}{(2 + i)} = \frac{10i + 5i^2}{4 + 2i - 2i - i^2} = \frac{10i + 5(-1)}{4 - (-1)} = \frac{-5 + 10i}{5} = \frac{-5}{5} + \frac{10i}{5} = -1 + 2i$$

Problem 11. Divide two complex numbers.

$$(a) \frac{-2+6i}{i} \quad \frac{-2i+6i^2}{i^2} \quad \frac{-6-2i}{-1} = \boxed{6+2i}$$

$$(b) \frac{5-2i}{-10i} \quad \frac{5i-2i^2}{-10i^2} \quad \frac{2+5i}{10} = \frac{2}{10} + \frac{5}{10}i = \boxed{\frac{1}{5} + \frac{1}{2}i}$$

$$(c) \frac{3+4i}{2+i} \quad \frac{(3+4i)(2-i)}{(2+i)(2-i)} \quad \frac{6-3i+8i-4i^2}{6+5i+4} = \frac{10+5i}{5} = \boxed{2+i}$$

$$(d) \frac{1-5i}{-2-3i} \quad \frac{(1-5i)(-2+3i)}{(-2-3i)(-2+3i)} \quad \frac{-2+3i+10i-15i^2}{-2+13i+15} = \frac{13+13i}{13} = \boxed{1+i}$$

Problem 12. * Perform the indicated operations.

$$(a) \sqrt{-16} \cdot \sqrt{-25} \quad 4i \cdot 5i = 20i^2 \quad 20(-1) = \boxed{-20}$$

$$(b) \sqrt{-14} \cdot \sqrt{7} \quad i\sqrt{14} \cdot \sqrt{7} = i\sqrt{98} = \boxed{7i\sqrt{2}}$$

$$(c) \frac{\sqrt{4}}{\sqrt{-9}} \quad \frac{2}{3i} \cdot \frac{i}{i} = \frac{2i}{3i^2} = \boxed{-\frac{2i}{3}}$$

$$(d) \frac{\sqrt{-40}}{\sqrt{-2}} \quad \frac{2i\sqrt{10}}{i\sqrt{2}} = \boxed{2\sqrt{5}}$$

$$(e) (\sqrt{-7} + 3) \cdot (2i) = (i\sqrt{7} + 3)(2i) = \frac{2i^2\sqrt{7} + 6i}{2(-1)} = \boxed{-2\sqrt{7} + 6i}$$

$$(f) \frac{1 + \sqrt{-9}}{2 - \sqrt{-4}} = \frac{(1 + 3i)(2 + 2i)}{(2 - 2i)(2 + 2i)}$$

$$\frac{2 + 2i + 6i + 6i^2}{2 + 8i - 6}$$

$$\frac{-4 + 8i}{8}$$

$$\frac{-1 + 2i}{2}$$

$$\boxed{-\frac{1}{2} + i}$$

Powers of i

Reminder

0
0.25 1
0.5 2
0.75 3

$$i^0 = 1$$

$$i^1 = i$$

$$i^2 = -1$$

$$i^3 = -i$$

$$i^0 = 1$$

$$i^1 = i$$

$$i^2 = -1$$

$$i^3 = i^2 \cdot i = (-1) \cdot i = -i$$

$$i^4 = i^2 \cdot i^2 = (-1)(-1) = 1$$

$$i^5 = i^4 \cdot i = 1 \cdot i = i$$

$$i^6 = i^4 \cdot i^2 = 1 \cdot (-1) = -1$$

$$i^7 = i^4 \cdot i^3 = 1 \cdot (-i) = -i$$

$$i^8 = i^4 \cdot i^4 = 1 \cdot 1 = 1 \text{ and so on.}$$

Powers of i cycle through the same four outcomes (i , -1 , $-i$, and 1) because i^4 has the same multiplicative property as 1. Also, any power of i with an exponent that is a multiple of 4 has value 1. As with real numbers, $i^0 = 1$.

Problem 13. Simplify each power of i .

$$(a) i^{80} = (i^4)^{20} = 1^{20} = \boxed{1}$$

$$\frac{80}{4} = 20 \text{ r } 0 \quad i^0 = \boxed{1}$$

$$(b) i^{14} = i^4 \cdot i^4 \cdot i^2 = 1 \cdot 1 \cdot (-1) = \boxed{-1}$$

$$(c) i^{33}$$

$$\frac{33}{4} = 8 \text{ r } 1 \quad i^1 = \boxed{i}$$

$$\frac{33}{4} = 8 \text{ r } 1$$

$$(d) i^{15} = (i^4)^3 \cdot i^3 = 1^3 \cdot (-i) = \boxed{-i}$$

$$(e) i^{-43}$$

$$\frac{1}{i^{43}} = \frac{1}{i^4 \cdot i^4 \cdot i^3} = \frac{1}{1 \cdot 1 \cdot (-i)} = \frac{1}{-i} = \frac{1}{-i} \cdot \frac{i}{i} = \frac{i}{-i^2} = \frac{i}{-(-1)} = \boxed{i}$$

$$(f) i^{-36}$$

$$\frac{1}{i^{36}} = \frac{1}{(i^4)^9} = \frac{1}{1^9} = \boxed{1}$$

$$\frac{i}{i} = \boxed{1}$$